

Block 1

Read, Explore, and Report on a Define-XML v2.1

15 min lecture • 20 min exercise • 5 min review

The one idea

define.xml	odmlib objects
<pre><ODM> L <Study> L <MetaDataVersion> <ItemGroupDef> L <ItemRef -> <ItemDef> ← L <CodeList></pre>	<pre>odm L .Study ← single object L .MetaDataVersion ← single object .ItemGroupDef [list] L .ItemRef [list] .ItemDef [list] L .CodeList [list]</pre>

Not a parse tree — a **typed object model of the v2.1 spec**.
Namespaces, XPath, and OID bookkeeping stop being your problem.

Three idioms — that's the whole block

```
# 1. load
loader = LD.ODMLoader(DL.XMLDefineLoader(model_package="define_2_1"))
loader.open_odm_document(path)
mdv = loader.root().Study.MetaDataVersion

# 2. navigate — collections are plain Python lists
mdv.ItemGroupDef · mdv.ItemDef · mdv.CodeList · mdv.MethodDef

# 3. find — searches descendants by class + attribute
mdv.find("ItemDef", "OID", "IT.DM.AGE")      # first match or None
mdv.find_all("ItemDef", "Name", "VSTESTCD")  # every match, a list
```

Element text is `_content` — `age.Description.TranslatedText[0]._content`

Beyond Define-XML

The same four lines, a different standard — swap the loader:

Standard	Loader	Note
Define-XML 2.0 / 2.1	<code>XMLDefineLoader</code>	today
ODM 1.3.2	<code>XMLODMLoader</code>	<code>Study[0]</code> , <code>MetaDataVersion[0]</code> — lists
ARM 1.0	<code>XMLArmLoader</code>	<code>mdv.AnalysisResultDisplays</code>
Dataset-JSON 1.1	<code>DatasetJSON.read_json()</code>	JSON — no XML loader needed

Learn the idioms once. **Block 1's stretch goal is the ARM row.**

Watch out for these

- Forgetting `model_package="define_2_1"` – v2.0 is the default
- Indexing `Study[0]` in Define-XML – single objects here, lists in ODM 1.3.2
- Calling `loader.root()` **repeatedly** – each call builds a *fresh* tree, so edits seem to vanish. Call once, keep the reference
- Reading `.text` for element content – odmlib uses `_content`
- Assuming every codelist has `CodeListItem` – enumerations use `EnumeratedItem` (coded values, no decodes). Check both

Your turn → `read_explore_exercise.ipynb` – 4 TODOs, ~20 min